

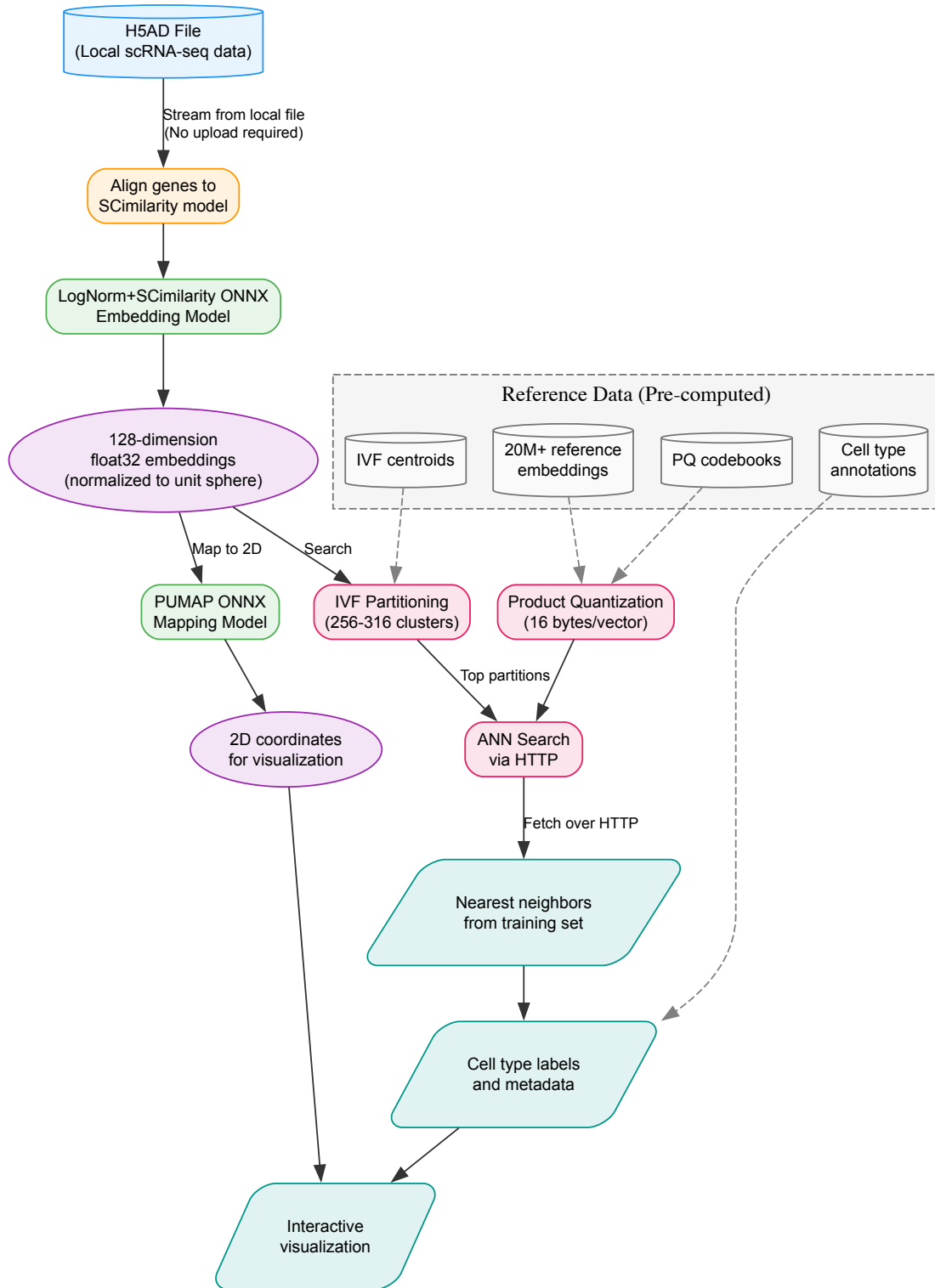
analysis

September 11, 2025

1 Cytoverse Analysis

CytoVerse is a browser-based platform for single-cell RNA-seq analysis, designed for cell annotation using foundation model embeddings. It operates entirely in the browser, streaming h5ad files from local storage without uploading data or requiring server computation. Here we compare its performance vs. the benchmark of SCimilarity running on a server. SCimilarity uses TileDB KNN IVF_FLAT index which provides exact vector similarity search. Cytoverse quantizes the vectors via Product Quantization (PQ) and then performs an approximate lookup based on partitions in an Inverted File Index (IVF) fetched over HTTP.

1.1 Data and Compute Flow



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3 Embeddings Concordance

100 randomly selected expression profiles from GSE136831_subsample_100.h5ad:
Max difference between SCimilarity (PyTorch) and ONNX: 3.31e-06

KNN execution time: 32.6146 seconds
KNN memory (~bandwidth) usage: 1056.35 MiB
KNN peak memory usage: 1056.35 MiB
KNN on-disk storage: 9023.19 MiB

IVF partitions: 0%| | 0/3 [00:00<?, ?it/s]
Probes (n_sub=8): 0%| | 0/8 [00:00<?, ?it/s]
Probes (n_sub=16): 0%| | 0/8 [00:00<?, ?it/s]
Probes (n_sub=32): 0%| | 0/8 [00:00<?, ?it/s]

3.1 Label Concordance

3.1.1 Comparative Analysis: Parameter Extremes

Metric	Lowest Parameters	Highest Parameters
	$n_{sub} = 8, n_{probe} = 1$	$n_{sub} = 32, n_{probe} = 32$
Mismatch Rate	47.0%	13.0%
Total Mismatches	47	13
Improvement	–	34.0 pp

3.1.2 Query Resolution Analysis

Category	Count
Queries fixed with higher parameters	40
New mismatches with higher parameters	6
Persistent mismatches	7

3.1.3 Top 5 Confusions Comparison

Lowest Parameters	Highest Parameters
5× astrocyte microglial cell	2× astrocyte microglial cell
4× glial cell radial glial ce	1× fibroblast luminal epithel
4× CD4-positive, a glial cell	1× myofibroblast c radial glial ce
2× microglial cell oligodendrocyte	1× glial cell microglial cell
2× macrophage microglial cell	1× enterocyte glial cell

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LaTeX code for Overleaf (copy and paste):

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% Table 1: Comparative Analysis
\begin{table}[h]
\centering
\caption{Comparative Analysis: Parameter Extremes}
\begin{tabular}{lcc}
\toprule
\textbf{Metric} & \textbf{Lowest Parameters} & \textbf{Highest Parameters} \\
& ($n_{\text{sub}}=8, n_{\text{probe}}=1$) & ($n_{\text{sub}}=32, n_{\text{probe}}=32$) \\
\midrule
Mismatch Rate & 47.0\% & 13.0\% \\
Total Mismatches & 47 & 13 \\
Improvement & -- & 34.0 pp \\
\bottomrule
\end{tabular}
\end{table}
```

```
% Table 2: Query Resolution Analysis
\begin{table}[h]
\centering
\caption{Query Resolution Analysis}
\begin{tabular}{lr}
\toprule
\textbf{Category} & \textbf{Count} \\
\midrule
Queries fixed with higher parameters & 40 \\
New mismatches with higher parameters & 6 \\
Persistent mismatches & 7 \\
\bottomrule
\end{tabular}
\end{table}
```

```
% Table 3: Top 5 Confusions Comparison
\begin{table}[h]
\centering
\caption{Top 5 Confusions Comparison}
\begin{tabular}{ll}
\toprule
\textbf{Lowest Parameters} & \textbf{Highest Parameters} \\
\midrule
5$\times$ astrocyte $\rightarrow$ microglial cell & 2$\times$ astrocyte $\rightarrow$ microglial cell \\
\end{tabular}
\end{table}
```

```

4$\times$ glial cell $\rightarrow$ radial glial ce & 1$\times$ fibroblast
$\rightarrow$ luminal epithel \\
4$\times$ CD4-positive, a $\rightarrow$ glial cell & 1$\times$ myofibroblast
c $\rightarrow$ radial glial ce \\
2$\times$ microglial cell $\rightarrow$ oligodendrocyte & 1$\times$ glial
cell $\rightarrow$ microglial cell \\
2$\times$ macrophage $\rightarrow$ microglial cell & 1$\times$ enterocyte
$\rightarrow$ glial cell \\
\bottomrule
\end{tabular}
\end{table}

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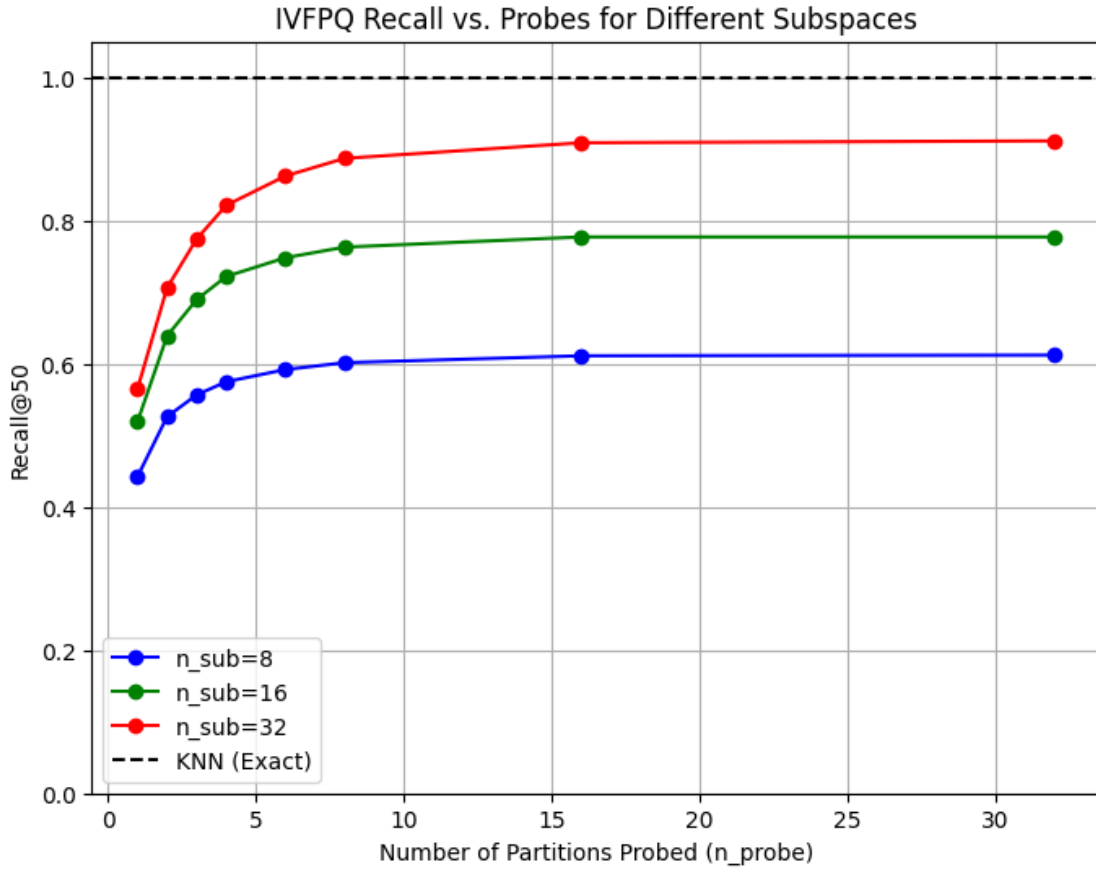
3.2 Performance

<pandas.io.formats.style.Styler at 0x5dbe043e0>

Method	n_sub	n_probe	Recall	Time (s)	Speedup	Storage	Reduction	BW
IVFPQ	8	1	0.442	1.8	17.9×	92 MB	98×	1.2 MB
IVFPQ	8	2	0.526	4.0	8.2×	92 MB	98×	2.3 MB
IVFPQ	8	3	0.556	5.8	5.6×	92 MB	98×	3.3 MB
IVFPQ	8	4	0.575	8.0	4.1×	92 MB	98×	4.4 MB
IVFPQ	8	6	0.592	12.1	2.7×	92 MB	98×	6.7 MB
IVFPQ	8	8	0.601	16.2	2.0×	92 MB	98×	8.4 MB
IVFPQ	8	16	0.611	39.3	0.8×	92 MB	98×	15.2 MB
IVFPQ	8	32	0.612	69.8	0.5×	92 MB	98×	26.0 MB
IVFPQ	16	1	0.519	1.8	18.6×	153 MB	59×	2.0 MB
IVFPQ	16	2	0.639	4.0	8.1×	153 MB	59×	3.7 MB
IVFPQ	16	3	0.689	5.9	5.5×	153 MB	59×	5.5 MB
IVFPQ	16	4	0.722	7.9	4.1×	153 MB	59×	7.4 MB
IVFPQ	16	6	0.748	13.2	2.5×	153 MB	59×	11.1 MB
IVFPQ	16	8	0.763	16.9	1.9×	153 MB	59×	14.1 MB
IVFPQ	16	16	0.777	35.0	0.9×	153 MB	59×	25.2 MB
IVFPQ	16	32	0.777	69.0	0.5×	153 MB	59×	43.4 MB
IVFPQ	32	1	0.565	1.9	17.5×	274 MB	33×	3.7 MB
IVFPQ	32	2	0.706	4.1	8.0×	274 MB	33×	6.8 MB
IVFPQ	32	3	0.774	6.0	5.5×	274 MB	33×	10.0 MB
IVFPQ	32	4	0.821	8.1	4.0×	274 MB	33×	13.3 MB
IVFPQ	32	6	0.862	12.3	2.7×	274 MB	33×	20.1 MB
IVFPQ	32	8	0.887	16.6	2.0×	274 MB	33×	25.5 MB
IVFPQ	32	16	0.908	35.5	0.9×	274 MB	33×	45.5 MB
IVFPQ	32	32	0.911	73.4	0.4×	274 MB	33×	77.9 MB
KNN	-	-	1.000	32.6	1.0×	9023 MB	1.0×	0.0 MB

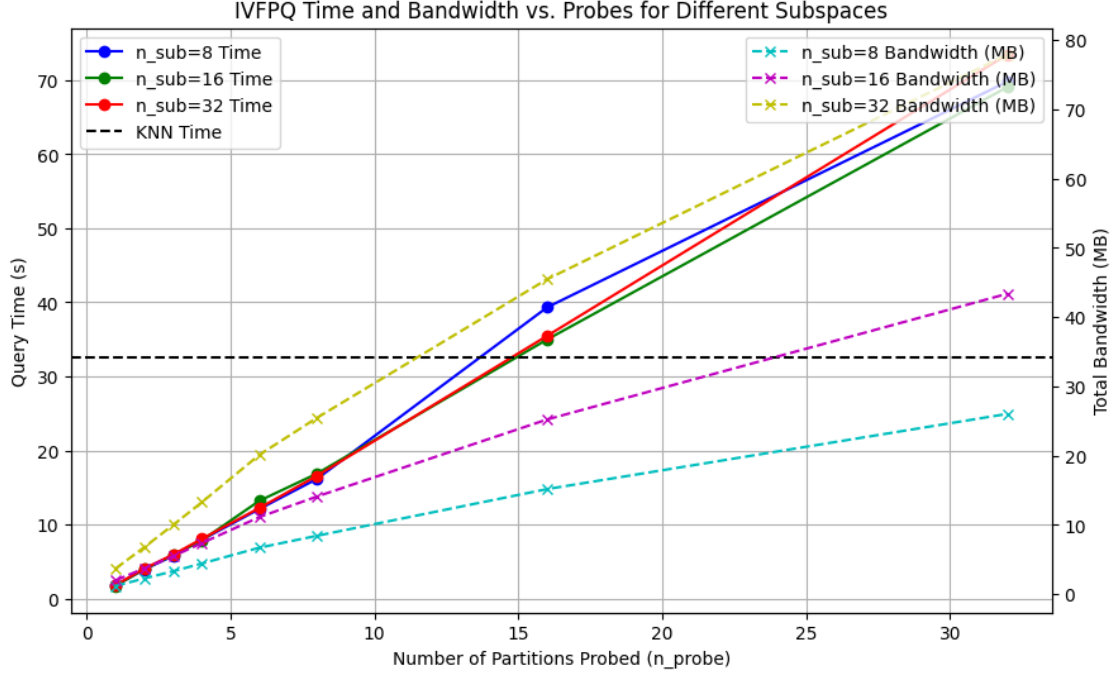
3.3 Recall vs. Probes

IVFPQ recall improves with more probes but plateaus. Fewer subspaces trade accuracy for compression reducing server storage and bandwidth.



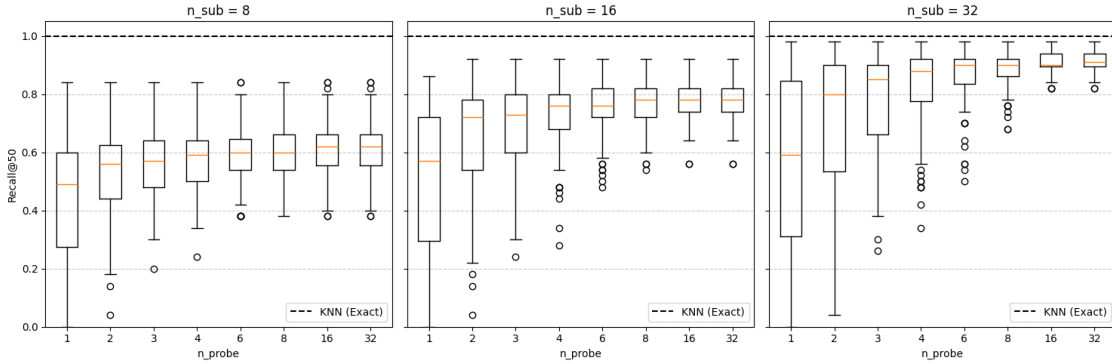
3.4 Time and Bandwidth vs. Probes

Shows tradeoff between PQ quantization and its impact on overall bandwidth and time vs. number of partitions probed. IVFPQ queries are faster than KNN, with bandwidth scaling linearly with probes



3.5 Recall@K by IVGPQ Parameter

Distribution of Recall@K across all queries for each n_probe value grouped by n_sub . Shows variability in recall (not just the mean) across queries, helping researchers assess consistency for single-cell RNA-seq labeling in a web app vs. local KNN.



3.6 Distance Distortion vs. Recall@K

Scatter plots of the relationship between mean distance distortion (for matched neighbors) and Recall@K for each query, with points colored by the average nearest neighbor distance in single-cell RNA-seq data (nn_dist_SC , reflecting query “difficulty”). This helps researchers understand if

queries with farther neighbors (harder cases) exhibit higher distortion or lower recall in IVFPQ compared to KNN, relevant for web app vs. local tool tradeoffs.

Distance Distortion vs. Recall@K (n_probe horizontal, n_sub vertical)

